The Korteweg–de Vries (KdV) Equation

The KdV Equation

Standard form:

$$u_t + 6uu_x + u_{xxx} = 0$$

Interpretation:

- $u(x, t)$: wave profile
- $6uu_x$: nonlinear steepening
- u_{xxx}: dispersive smoothing

Context:

- Shallow water waves
- Balance of nonlinearity and dispersion

Conservation Laws

The KdV equation has an infinite number of conserved quantities.

First three:

$$M = \int_{\mathbb{R}} u \, dx$$

$$P = \int_{\mathbb{R}} u^2 \, dx$$

$$E = \int_{\mathbb{R}} \left(\tfrac{1}{2} u_x^2 - u^3\right) dx$$

We focus on the energy functional:

$$E[u] = \int \left(\tfrac{1}{2} u_x^2 - u^3\right) dx$$

Energy Conservation

Compute the time derivative:

$$\frac{dE}{dt} = \int \left(u_x u_{xt} - 3u^2 u_t \right) dx$$

Use the PDE:

$$u_t = -6uu_x - u_{xxx}$$

Substitute and integrate by parts (assuming decay at infinity).

Result:

$$\frac{dE}{dt} = 0$$

Conclusion: Energy is conserved.

Sketch of Energy Derivation

Key steps:

- Replace u_t using KdV
- Terms become total derivatives:

$$\int u_x u_{xxxx}\,dx, \quad \int u^2 u_{xxx}\,dx$$

- Integrate by parts repeatedly
- Boundary terms vanish (rapid decay or periodic BCs)

Takeaway:

$$\frac{d}{dt}\int \left(\tfrac{1}{2}u_x^2 - u^3\right)dx = 0$$

Existence and Uniqueness

Theorem (Local Well-Posedness):
Let $s > \frac{3}{2}$ and $u_0 \in H^s(\mathbb{R})$. Then there exists $T > 0$ and a unique solution

$$u \in C([0, T]; H^s(\mathbb{R}))$$

to the KdV equation.

Sketch of proof:

- Rewrite as evolution equation:

$$u_t = -6uu_x - u_{xxx}$$

- Linear part generates a unitary group
- Use fixed point argument (Picard iteration)
- Sobolev embedding controls uu_x

Global Existence

Theorem (Global existence):
For sufficiently smooth initial data, solutions exist for all $t > 0$.

Idea:

- Use conservation of $\|u\|_{L^2}$
- Use energy conservation
- Prevent blow-up via a priori bounds

KdV is completely integrable, giving strong global control.

Travelling Wave Ansatz

Seek solutions of the form:

$$u(x,t) = f(x - ct)$$

Let $\xi = x - ct$.

Then:

$$u_t = -cf', \quad u_x = f', \quad u_{xxx} = f'''$$

Substitute into KdV:

$$-cf' + 6ff' + f''' = 0$$

Reduction to ODE

Integrate once:

$$-cf + 3f^2 + f'' = A$$

For localized solutions, take $A = 0$:

$$f'' = cf - 3f^2$$

Multiply by f' and integrate:

$$\frac{1}{2}(f')^2 = \frac{c}{2}f^2 - f^3$$

Soliton Solution

Solve:

$$(f')^2 = cf^2 - 2f^3$$

This yields:

$$f(\xi) = \frac{c}{2}\,\mathrm{sech}^2\left(\frac{\sqrt{c}}{2}\xi\right)$$

Final travelling wave:

$$u(x,t) = \frac{c}{2}\,\mathrm{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct)\right)$$

Properties:

- Localized
- Stable
- Shape preserved under interaction

Summary

- KdV equation:

$$u_t + 6uu_x + u_{xxx} = 0$$

- Conserved energy:

$$E = \int \left(\tfrac{1}{2}u_x^2 - u^3\right) dx$$

- Well-posed in Sobolev spaces
- Admits soliton solutions

Second Conserved Quantity

We derive conservation of

$$P[u] = \int_{\mathbb{R}} u^2 \, dx$$

Compute:

$$\frac{d}{dt} \int u^2 \, dx = 2 \int u u_t \, dx$$

Use the KdV equation:

$$u_t = -6 u u_x - u_{xxx}$$

Thus:

$$\frac{d}{dt} \int u^2 dx = -12 \int u^2 u_x \, dx - 2 \int u u_{xxx} \, dx$$

Derivation (Integration by Parts)

We handle each term separately.

First term:

$$\int u^2 u_x \, dx = \frac{1}{3} \int (u^3)_x \, dx = 0$$

Second term:

$$\int u u_{xxx} \, dx = -\int u_x u_{xx} \, dx = -\frac{1}{2} \int (u_x^2)_x \, dx = 0$$

(All boundary terms vanish for decaying or periodic solutions.)

Conclusion:

$$\frac{d}{dt} \int u^2 \, dx = 0$$

So: $P[u] = \int u^2 dx$ is conserved.

Hierarchy of Conserved Quantities

The KdV equation has an infinite sequence of conserved quantities.

First three:

$$M[u] = \int_{\mathbb{R}} u\,dx \quad \text{(mass)}$$
$$P[u] = \int_{\mathbb{R}} u^2\,dx \quad \text{(momentum)}$$
$$E[u] = \int_{\mathbb{R}} \left(\tfrac{1}{2}u_x^2 - u^3\right) dx \quad \text{(energy)}$$

Structure:

- Each conserved quantity controls higher regularity
- They arise from an underlying structure structure

Key takeaway: KdV has infinitely many conservation laws

This is a hallmark of *complete integrability*.

Lax Pair Formulation

A key feature of KdV is that it admits a **Lax pair**.

Define operators:

$$L = -\partial_x^2 + u(x,t), \qquad P = -4\partial_x^3 + 3(u\partial_x + \partial_x u)$$

Then the KdV equation is equivalent to:

$$\frac{dL}{dt} = [P, L] = PL - LP$$

Interpretation:

- Evolution is isospectral
- Eigenvalues of L are time-invariant

Verification of the Lax Pair

Apply both sides to a test function ψ.

Left-hand side:

$$\frac{d}{dt}(L\psi) = u_t\psi + L\psi_t$$

Right-hand side:

$$[P, L]\psi = P(L\psi) - L(P\psi)$$

A direct computation shows:

$$[P, L] = (6uu_x + u_{xxx}) \cdot (\text{multiplication operator})$$

Thus:

$$\frac{dL}{dt} = [P, L] \quad \Longleftrightarrow \quad u_t + 6uu_x + u_{xxx} = 0$$

Spectral Problem

Consider the Schrödinger operator:

$$L\psi = \lambda\psi \quad \Longleftrightarrow \quad -\psi'' + u(x,t)\psi = \lambda\psi$$

Key idea:

- $u(x,t)$ acts as a potential
- The spectrum of L encodes the solution

Under KdV flow:

$$\lambda \text{ is constant in time}$$

This is the basis of inverse scattering.

Direct Scattering

Given initial data $u(x, 0)$:

Solve:

$$-\psi'' + u(x, 0)\psi = \lambda\psi$$

Extract **scattering data**:

- Reflection coefficient $R(\lambda)$
- Discrete eigenvalues $\{\lambda_n\}$
- Norming constants

Interpretation:

- Continuous spectrum $\rightarrow$ radiation
- Discrete spectrum $\rightarrow$ solitons

Time Evolution of Scattering Data

Under KdV evolution:

- Eigenvalues λ_n are constant
- Reflection coefficient evolves simply:

$$R(\lambda, t) = R(\lambda, 0)\, e^{8i\lambda^{3/2} t}$$

Key simplification:

- Complicated PDE $\rightarrow$ simple phase evolution

Inverse Scattering

Reconstruct $u(x, t)$ from scattering data.

Steps:

1. Compute scattering data from $u(x, 0)$
2. Evolve data explicitly in time
3. Solve inverse problem (Gel'fand–Levitan–Marchenko equation)

Outcome:

- Exact solution for all t
- Solitons emerge from discrete spectrum

Solitons and the Spectrum

Each negative eigenvalue corresponds to a soliton.

For a single eigenvalue:

$$\lambda = -\kappa^2$$

This produces:

$$u(x,t) = \frac{c}{2}\operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct)\right)$$

Interpretation:

- Discrete spectrum $\leftrightarrow$ solitons
- Multi-soliton solutions from multiple eigenvalues

Summary: Integrability

- Lax pair $\Rightarrow$ isospectral flow
- Infinite conserved quantities
- Scattering data evolves linearly
- Nonlinear PDE solved via linear spectral theory

Conclusion:

KdV is a completely integrable system

Nonlinear Schrödinger Equation (NLS)

Standard (focusing) form:

$$i\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0$$

Interpretation:

- $\psi(x, t)$: complex wave envelope
- ψ_{xx}: dispersion
- $2|\psi|^2\psi$: nonlinear self-phase modulation

Connection to KdV:

- Both are completely integrable
- Both admit soliton solutions
- Both arise from balance of nonlinearity and dispersion

NLS in Optical Communication

Physical setting:

- Light propagation in optical fibers
- $\psi(x, t)$ represents a slowly varying electric field envelope

Key effects:

- Dispersion spreads pulses
- Nonlinearity focuses them

Solitons:

- Stable pulse solutions
- Maintain shape over long distances
- Resist dispersion-induced broadening

Application:

- Soliton-based communication systems
- Long-distance transmission with minimal distortion